FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY

MICROWAVE DEVICES AND RADAR

MODEL QUESTION PAPER

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	Name an example where TEM mode of propagation is possible?	M1.04	R
2	Define coupling factor of directional coupler?	M1.04	R
3	Show the S-matrix of an isolator?	M1.03	R
4	Define relation between gain and directivity of an antenna?	M2.03	R
5	Draw and label a parabolic reflector antenna?	M2.04	R
6	Write any 2 applications of Gunn diode?	M3.04	R
7	Draw the structure of microwave TUNNEL diode?	M3.01	R
8	What is the use of duplexer in a RADAR?	M4.01	R
9	What is PRF in a RADAR?	M4.02	R

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Show the S-matrix of magic tee?	M1.03	R
2	Show the S-matrix of a directional coupler?	M1.03	R
3	List three microwave frequency bands with its applications	M1.01	R
4	Recall the operation of magnetron?	M2.02	R
5	Define gain and directivity of an antenna?	M2.03	R

6	List applications of microwave tunnel diode	M3.02	R
7	Draw and explain the VI characteristics of gunn diode?	M3.04	U
8	Recall the working principle of microwave BJT?	M3.01	R
9	List and explain 3 applications of RADAR?	M4.02	R
10	Draw the block diagram of MTI Radar with power amplifier?	M4.04	R

PART C

III. Answer all questions from the following

(6x 7 = 42 Marks)

		Module Outcome	Cognitive level
1	Summarize different propagation modes in rectangular waveguide?	M1.04	U
OR			
2	Explain the significance of each components in the S-matrix of a 2-port device?	M1.03	U
3	Explain three basic parameters of dipole antenna with figure if necessary?	M2.04	U
OR			
4	Summarize the working of horn antenna and parabolic reflector antenna?	M2.04	U
5	Compare the structure and working of klystron and reflex klystron?	M2.01	U
OR			
6	Explain the operating principle of travelling wave tube amplifier with suitable diagram	M2.02	U
7	Summarize the construction and operation of microwave Tunnel diode?	M3.01	U
OR			
8	Explain the working principle of PIN diode with suitable diagram?	M3.03	U
9	Derive the RADAR range equation?	M4.03	U
OR			
10	a) Explain the working of FM - CW Radar with suitable block diagram? b) Explain doppler effect?	M4.05	U
11	Summarize the working of MTI RADAR with suitable block diagram?	M4.04	U
OR			
12	Explain Radar performance factors and its significance?	M4.02	U

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PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	List transmission modes in a rectangular waveguide?	M1.04	R
2	Define directivity of directional coupler?	M1.04	R
3	Name three antennas?	M2.04	R
4	Define gain of an antenna?	M2.03	R
5	List two applications of Gunn diode?	M3.04	R
6	Show the structure of PIN diode?	M3.03	R
7	List two applications of RADAR?	M4.02	R
8	What is the expansion for acronym RADAR	M4.01	R
9	Define doppler effect?	M4.04	R

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Define different propagation modes in rectangular waveguide?	M1.04	R
2	Relate reflection coefficient and transmission coefficient?	M1.03	R
3	Define each term in the S-matrix of a 2-port isolator?	M1.03	R
4	Classify microwave frequency range?	M1.01	R
5	Relate directivity and gain of an antenna?	M2.03	R
6	Draw the diagram and label important parts of reflex klystron?	M2.01	R
7	Draw the structure of PIN diode and explain its working?	M3.03	U
8	List three difference of microwave BJT with normal BJT?	M3.01	R
9	What are the following terms. a) Radar cross section of targets and b) pulse repetition frequency.	M4.02	R
10	List the applications of Tunnel diode?	M3.02	R

PART C

III. Answer all questions from the following

(6x 7 = 42 Marks)

		Module Outcome	Cognitive level
1	Explain the working of magic Tee with the help of its S-matrix?	M1.03	U
OR			
2	Compare the S-matrix of circulator and a 3-port isolator.	M1.03	U
3	a) Summarize three basic parameters of an antenna? b) Explain these parameters for dipole antenna with figures if necessary?	M2.03	U
OR			
4	Illustrate the working of travelling wave tube amplifier with suitable diagram?	M2.02	U
5	Explain the working principle of Magnetron with figure?	M2.02	U
OR			
6	Explain the working of Klystron with suitable diagram?	M2.01	U
7	Explain Gunn effect with the help of VI characteristics of gunn diode?	M3.04	U
OR			
8	Explain the structure and working of Tunnel diode?	M3.01	U
9	Derive the RADAR range equation?	M4.03	U
OR			
10	Explain 7 applications of RADAR?	M4.02	U
11	Explain the block diagram of FM - CW Radar with super heterodyne receiver?	M4.05	U
OR			
12	Summarize the working of MTI RADAR with suitable block diagram?	M4.04	U